

MCALESTER, OK, USA

WMO: 723566

Lat: 34.882N

Lon: 95.783W

Elev: 235

StdP: 98.54

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.4	-15.3	1.0	-5.6	-12.7	1.3	-3.6	11.1	9.8	10.0	9.1	3.2	0	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	37.8	23.9	36.1	24.0	34.4	24.1	26.2	33.0	25.6	32.4	25.1	31.8	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.7	29.7	23.7	19.1	29.1	23.1	18.4	28.5	82.4	33.0	80.0	32.5	77.7	31.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.4	7.6	-13.3	39.3	2.8	2.2	-15.3	40.9	-16.9	42.1	-18.5	43.4	-20.5	45.0		
(5)			-13.8	27.3	2.6	0.9	-15.7	28.0	-17.3	28.5	-18.8	29.1	-20.7	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	5.2	7.5	12.2	16.6	21.1	25.7	28.0	27.8	23.7	17.5	11.5	6.4	(6)
(7)		DBStd	9.30	6.05	6.19	5.62	4.56	3.89	2.75	2.69	2.87	4.02	4.92	5.35	5.57	(7)
(8)		HDD10.0	520	172	111	42	5	1	0	0	0	0	4	46	139	(8)
(9)		HDD18.3	1684	407	306	202	84	19	0	0	0	7	76	213	371	(9)
(10)		CDD10.0	3071	23	40	109	204	346	471	559	553	411	237	90	28	(10)
(11)		CDD18.3	1193	1	2	11	33	106	221	301	294	167	51	7	1	(11)
(12)		CDH23.3	12762	2	16	78	252	877	2281	3611	3477	1693	437	39	2	(12)
(13)		CDH26.7	5877	0	4	11	41	238	972	1884	1841	758	124	5	0	(13)
(14)	Wind	WSAvg	3.2	3.7	3.8	4.0	4.0	3.3	2.9	2.6	2.4	2.5	3.0	3.4	3.4	(14)
(15)	Precipitation	PrecAvg	1116	52	67	96	105	144	112	60	74	122	112	86	73	(15)
(16)		PrecMax	1686	161	185	226	273	318	225	164	242	318	443	249	212	(16)
(17)		PrecMin	526	0	7	24	5	21	13	0	1	7	3	12	5	(17)
(18)		PrecStd	293	38	42	53	62	77	57	47	55	82	104	58	54	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	25.1	28.2	29.9	32.6	35.9	39.9	40.1	37.7	32.0	27.1	23.1	(19)
(20)			MCWB	14.1	15.5	18.1	19.1	23.4	24.3	23.5	23.5	22.9	21.4	18.4	17.1	(20)
(21)		2%	DB	20.2	22.2	25.4	27.7	30.8	34.1	37.8	38.0	34.8	30.0	24.2	20.6	(21)
(22)			MCWB	14.4	15.0	16.9	19.0	22.7	24.5	24.0	23.8	23.0	20.4	16.8	15.2	(22)
(23)		5%	DB	17.9	19.9	23.4	26.2	29.2	32.8	36.0	36.4	32.9	27.8	22.2	18.0	(23)
(24)			MCWB	13.1	14.3	16.0	18.1	21.9	24.1	24.4	23.8	22.6	19.6	16.6	13.6	(24)
(25)		10%	DB	15.6	17.6	21.3	24.6	27.8	31.6	34.1	34.5	31.1	25.8	20.2	15.8	(25)
(26)			MCWB	10.5	12.9	14.9	17.4	21.4	23.9	24.5	23.9	22.1	18.9	15.2	11.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	18.6	19.8	21.9	25.2	26.9	26.9	26.6	25.3	23.6	20.8	18.8	(27)
(28)			MCD B	20.4	22.0	25.8	26.6	30.6	32.9	33.5	33.4	32.3	28.5	24.5	22.1	(28)
(29)		2%	WB	16.2	17.3	18.5	20.8	24.0	25.8	26.3	25.8	24.5	22.6	19.0	16.7	(29)
(30)			MCD B	18.9	20.0	23.2	25.3	28.8	31.9	33.2	33.0	31.2	27.1	22.2	19.6	(30)
(31)		5%	WB	13.9	15.3	17.4	20.1	23.2	25.2	25.7	25.2	23.8	21.6	17.6	14.8	(31)
(32)			MCD B	16.8	19.1	21.8	24.2	27.8	31.1	32.6	32.6	30.1	25.4	20.8	17.4	(32)
(33)		10%	WB	11.0	13.0	16.1	19.0	22.4	24.5	25.2	24.8	23.2	20.4	16.1	12.2	(33)
(34)			MCD B	14.7	16.9	20.6	23.2	26.8	29.9	32.0	31.8	28.8	24.2	19.8	14.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	11.5	11.8	12.0	10.4	10.3	11.0	11.6	12.0	12.3	11.8	10.9	(35)
(36)			MCDBR	14.8	14.5	13.7	13.1	11.4	11.4	13.2	14.1	13.8	13.9	13.2	13.8	(36)
(37)			MCWBR	9.5	9.3	7.5	6.8	5.5	4.5	3.9	4.1	4.8	6.4	7.8	9.4	(37)
(38)		5% WB	MCD BR	11.6	12.3	11.5	10.0	9.6	10.1	10.9	11.3	11.0	10.0	11.3	11.4	(38)
(39)			MCWBR	9.4	9.3	7.6	6.7	5.3	4.7	4.3	4.1	4.3	5.6	8.6	9.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.315	0.349	0.384	0.417	0.427	0.443	0.441	0.394	0.348	0.322	0.301	(40)	
(41)		taud	2.494	2.493	2.422	2.332	2.302	2.307	2.306	2.284	2.377	2.493	2.510	2.525	(41)	
(42)		Ebn at Noon	902	934	923	901	871	858	842	837	865	883	876	888	(42)	
(43)		Edh at Noon	84	93	109	126	131	130	130	130	113	93	82	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.63	3.12	4.18	5.14	5.63	6.43	6.54	5.98	4.98	3.80	2.80	2.25	(44)	
(45)		RadStd	0.27	0.46	0.28	0.35	0.55	0.71	0.39	0.39	0.46	0.54	0.35	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+92
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76

Nomenclature: See separate page